

The PI3K-Mediated Activation of CRAC Independently Regulates Adenylyl Cyclase Activation and Chemotaxis

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Summary

The ability of a cell to detect an external chemical signal and initiate a program of directed migration along a gradient comprises the fundamental process called chemotaxis [1]. Investigations in *Dictyostelium discoideum* and neutrophils have established that pleckstrin homology (PH) domain-containing proteins that bind to the PI3K products PI(3,4)P₂ and PI(3,4,5)P₃, such as CRAC (cytosolic regulator of adenylyl cyclase) and Akt/PKB, translocate specifically to the leading edge of chemotaxing cells [2–4]. CRAC is essential for the chemoattractant-mediated activation of the adenylyl cyclase ACA [5], which converts ATP into cAMP, the primary chemoattractant for *D. discoideum*. The mechanisms by which CRAC activates ACA remain to be determined. We now show that in addition to its essential role in the activation of ACA, CRAC is involved in regulating chemotaxis. Through mutagenesis, we show that these two functions are independently regulated downstream of PI3K. A CRAC mutant that has lost the capacity to bind PI3K products does not support chemotaxis and shows minimal ACA activation. Finally, overexpression of CRAC and various CRAC mutants show strong effects on ACA activation with little effect on chemotaxis. These findings establish that chemoattractant-mediated activation of PI3K is important for the CRAC-dependent regulation of both chemotaxis and adenylyl cyclase activation.

Results and Discussion

CRAC Regulates Chemotaxis

To assess whether CRAC (cytosolic regulator of adenylyl cyclase) is required for chemotaxis, cells lacking CRAC (*crac*[−]) [5] were starved and allowed to differentiate to the chemotaxis competent stage [2], as judged by Western blot analysis of early developmental markers, such as the chemoattractant receptor cAR1 (Figure S1 in the Supplemental Data available with this article online). The chemotactic behavior of a large population of cells was measured by the under-agarose chemotaxis assay. We observed that, in contrast to wild-type Ax3 cells, *crac*[−] cells do not move toward the chemoattractant source,

even after extended exposure to the chemoattractant (Figure 1A). To characterize this behavior at the cellular level, we visualized how cells respond to a point source of chemoattractant that is delivered with a micropipette. Here, again, we found that *crac*[−] cells show reduced chemotaxis. Within minutes after being exposed to the gradient, wild-type Ax3 cells rapidly align and migrate to the tip of the micropipette. Cells lacking CRAC, on the other hand, show much slower movement, barely moving one cell length during the 12 min recording, compared to >4 cell length for wild-type cells (Figure 1B, Movies 1 and 2, and Figure S2 for DIAS analysis). Although the *crac*[−] cells do eventually polarize and orient toward the tip of the micropipette, they display a broad front and do not form streams of cells aligned in a head-to-tail fashion, compared to wild-type cells.

To investigate whether the chemotaxis defect of the *crac*[−] cells is due to a fundamental defect in the mobilization of actin, we measured chemoattractant-dependent actin polymerization. We observed that, like wild-type cells, *crac*[−] cells show a rapid peak of cortical F-actin accumulation (within 5 s) upon stimulation (Figure 1C), demonstrating that *crac*[−] cells can mobilize the actin cytoskeleton in a chemoattractant-dependent manner. Indeed, in agreement with previous work showing that *crac*[−] cells exhibit normal orientation when placed in a chemoattractant gradient, we also find that *crac*[−] cells are able to polarize in response to chemoattractant stimulation [6] (Figure 1B and Movie 2). Interestingly, when we measure the extent by which chemoattractants can polymerize actin as a function of time, we find that *crac*[−] cells do not display the second phase of actin polymerization at approximately 120 s poststimulation, which has been proposed to play a role in pseudopod extension and chemotaxis (Figure 1D) [7]. Taken together, these findings show that the chemotaxis defects of *crac*[−] cells are not due to alterations in the motility machinery but, instead, to defects in translating the external chemoattractant gradient into directed pseudopod extension and subsequent directed migration. Furthermore, these results establish that, similar to RasC, AleA (a RasGEF), and the cytosolic regulator Pianissimo, CRAC has at least two functions [8–11]. Namely, in addition to its established role in adenylyl cyclase activation, CRAC is also involved in regulating chemotaxis.

CRAC Regulates Adenylyl Cyclase and Chemotaxis Independently

In order to begin to understand the mechanisms by which CRAC regulates the activation of both adenylyl cyclase and chemotaxis, we generated and characterized a series of C-terminal deletion mutants (Δ C-CRAC, Figure 2A), including the *synag7* deletion mutant, which was previously isolated from a chemical mutagenesis screen and shown to be defective in adenylyl cyclase activation [12]. The various CRAC mutants were fused to either HA (at the N terminus) or GFP (at the C terminus)

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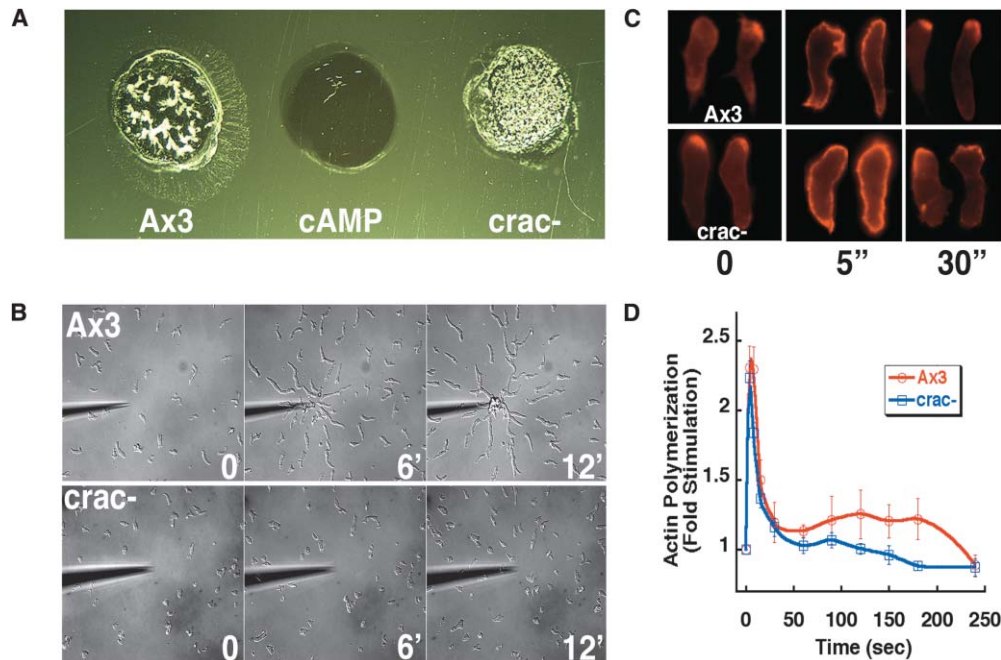


Figure 1. CRAC Regulates Chemotaxis

(A) Under-agarose chemotaxis assay. Differentiated wild-type Ax3 and *crac*⁻ cells were plated into wells on either side of a well containing 1 μ M cAMP. Images were taken with a Leica DM IL stereoscope 45 min after the cells were plated. Chemotaxing cells exit from the well and migrate under the agarose, appearing as a halo outside the perimeter of the well. Data are representative of results obtained from at least 6 independent experiments, each performed in triplicate.

(B) Chemotaxis to a point source of chemoattractant. Images of differentiated wild-type Ax3 (top panels and Movie 1) and *crac*⁻ cells (bottom panels and Movie 2) migrating to a micropipette filled with 10 μ M cAMP are shown. We could not measure any significant movement from the *crac*⁻ cells in the presence of 1 μ M cAMP, although wild-type cells responded very well to this concentration. Images representative of at least five independent experiments are shown.

(C and D) Actin polymerization response in response to chemoattractant stimulation. (C) Differentiated cells were plated onto 8-well chambered coverslips, stimulated with 10 μ M cAMP, fixed at the indicated time points, and then F-actin was visualized by Texas-Red-X phalloidin. Montages of two representative cells presented for each time point. (D) Differentiated cells were stimulated with 10 μ M cAMP and, at the indicated time points, aliquots were removed and fixed in potassium phosphate-PIPES buffer containing TRITC-phalloidin. Labeled actin was extracted with MeOH and fluorescence intensity was measured in a spectrofluorimeter. SEM values are presented from four independent experiments.

and expressed in *crac*⁻ cells. The resulting cell lines were allowed to differentiate and were processed through a battery of biological and biochemical analyses. We found that, like *crac*⁻ cells, all of the Δ C-CRAC mutants fail to aggregate when plated on non-nutrient agar (data not shown) and are defective in chemoattractant-mediated activation of adenylyl cyclase (Figure 2B). Nevertheless, the Δ C-CRAC mutants can translocate to the plasma membrane when stimulated with a uniform, saturating dose of chemoattractant, as judged by fluorescence microscopy and Western blot analysis of membrane preparations (Figure 2C). In addition, all of the Δ C-CRAC mutants are capable of rescuing the *crac*⁻ chemotaxis defect, as judged by both the micropipette and the under-agarose assays (Figure 2D). Cells that lack adenylyl cyclase can chemotax but cannot relay the chemoattractant signal to distal cells. As a result, they do not organize in head to tail fashion in long streams of cells during chemotaxis [13, 14]. Likewise, the Δ C-CRAC mutant-expressing cells, which are defective in the capacity to activate adenylyl cyclase, do not stream during chemotaxis (Figure 2D and Movie 3). From these studies, we conclude that the determinants under-

lying CRAC's role during chemotaxis are distinct from those involved in regulating adenylyl cyclase activation.

PI3K Products Regulate Chemotaxis and Adenylyl Cyclase Activation

The PH (pleckstrin homology) domain of CRAC can bind, in vitro, to the 3-phosphoinositides PI(3,4,5)P₃ and PI(3,4)P₂, the lipid products of PI3K [15]. We thus characterized a series of CRAC PH domain constructs in order to assess the role of PI3K in chemoattractant-mediated CRAC signaling. Here again the various PH mutants were fused to GFP to allow microscopic characterization of the translocation. We found that, similar to other PH domains [1], the CRAC PH domain alone (PH_{CRAC}) expressed in *crac*⁻ cells is sufficient to mediate membrane targeting upon either a uniform, saturating dose of chemoattractant or in a gradient of chemoattractant supplied by a micropipette (Figure S3). In spite of its proper localization in response to chemoattractant, PH_{CRAC} does not rescue the chemotaxis defect of *crac*⁻ cells (Figure S3) and *crac*⁻ cells expressing PH_{CRAC} have no detectable adenylyl cyclase activation following chemoattractant stimulation (data not shown). These results demon-

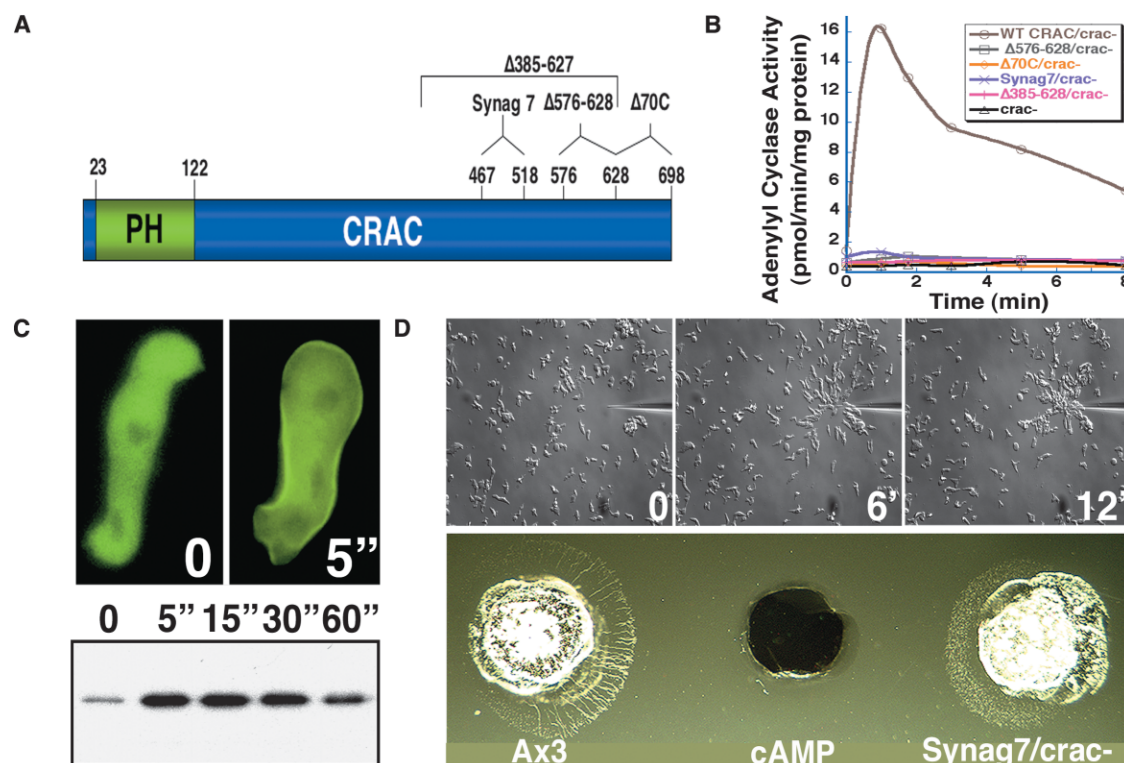


Figure 2. CRAC Regulates Adenylyl Cyclase and Chemotaxis Independently

(A) Schematic representation of CRAC C-terminal deletion (Δ C-CRAC) mutants.

(B) Δ C-CRAC mutants do not activate adenylyl cyclase. Adenylyl cyclase activity was measured after the addition of 10 μ M cAMP in *crac*⁻ cells expressing either wild-type CRAC or Δ C-CRAC mutants. Data are representative of results from three to four independent experiments, each performed in duplicate.

(C) Δ C-CRAC mutants translocate to the plasma membrane upon chemoattractant stimulation. In the top panels, differentiated cells were stimulated with 10 μ M cAMP and visualized microscopically, either live or after fixation 5 s poststimulation. For the biochemical translocation assay (bottom panel), cells were stimulated with cAMP and, at the indicated time points, aliquots were removed, diluted with cold buffer, and the membrane fractions were collected by centrifugation. Membrane fractions were analyzed by immunoblotting with either anti-GFP or anti-HA antibodies, depending on the construct. Results shown are for the *synag7* deletion mutant and are representative of all of the Δ C-CRAC mutants tested.

(D) Δ C-CRAC mutants can support chemotaxis. Chemotaxis of *crac*⁻ cells expressing Δ C-CRAC mutants was assessed by both the micropipette assay (top panel and Movie 3) and the under-agarose assay (bottom panel) as described in the legend of Figure 1. The Δ C-CRAC mutants show chemotaxis in both assays but do not organize in a head-to-tail fashion in long streams as wild-type cells do. Results shown are for the *synag7* deletion mutant and are representative of all of the Δ C-CRAC mutants tested.

strate that, although the PH domain is responsible for CRAC's recruitment to the plasma membrane, it does not contain the information required to mediate chemotaxis or to activate adenylyl cyclase on its own. Moreover, because we observed normal translocation kinetics, these findings show that the regulation of the PI3K/PTEN signaling cascade is intact in *crac*⁻ cells.

To further test the role of the PH domain in CRAC function, we generated a mutant with a deletion of the entire PH domain (Δ PH-CRAC) and expressed it in *crac*⁻ cells. We were unable to detect chemoattractant-mediated recruitment of Δ PH-CRAC to the plasma membrane (Figure S4), further confirming that the PH domain alone is a primary determinant of CRAC recruitment to the plasma membrane. Similarly, Δ PH-CRAC did not rescue either the chemotaxis defect (Figure S4) or the adenylyl cyclase defect of *crac*⁻ cells (Figure 3C). These results show that the PH domain of CRAC is essential for all of its known chemoattractant-stimulated functions, namely, membrane recruitment, chemotaxis, and adenylyl cyclase activation.

To examine more specifically the role of 3-phosphoinositides in CRAC function, we introduced a point mutation into the predicted 3-phosphoinositide binding motif [16] within the CRAC PH domain (R42C-CRAC). We demonstrate that this point mutant fails to bind to 3-phosphoinositides in vitro (Figure 3A). As with Δ PH-CRAC, we were unable to detect cAMP stimulated translocation of R42C-CRAC to the plasma membrane, either by fluorescence microscopy or Western blot analysis of membrane fractions (Figure 3B). In addition, the R42C-CRAC point mutant failed to support chemotaxis in either the micropipette assay (Figure 3B and Movie 4) or the under-agarose assay (Figure 3B), even after extended exposure to the chemoattractant gradient. This finding demonstrates that 3-phosphoinositide binding is essential for CRAC-mediated regulation of chemotaxis. Interestingly, we also find that the R42C-CRAC point mutant only partially rescues the adenylyl cyclase defects of *crac*⁻ cells (Figure 3C), suggesting that 3-phosphoinositide binding is important for CRAC-mediated activation of adenylyl cyclase. These findings have two implica-

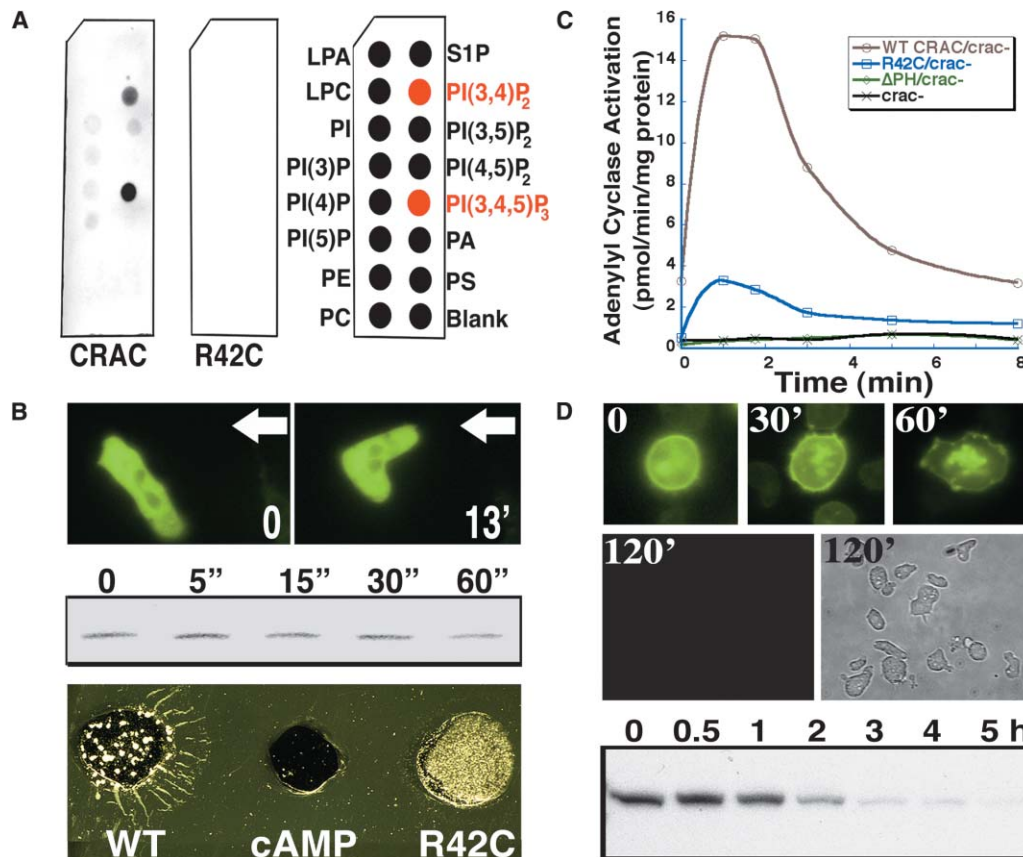


Figure 3. PI3K Products Regulate Chemotaxis and Adenylyl Cyclase Activation

(A) Dot blot assays for phosphoinositide binding. Supernatants of *crac*⁻ cells expressing GFP fusions of either wild-type CRAC (left panel) or R42C-CRAC (right panel) were incubated with dot blots containing a variety of lipids, as indicated in the key to the right. Both supernatants expressed comparable levels of protein, as judged by Western analysis. Bound proteins were detected with anti-GFP antibody.

(B) Chemotaxis and translocation of R42C-CRAC/*crac*⁻ cells. The R42C-CRAC mutant fails to support chemotaxis in either the micropipette assay (top panels and Movie 4) or the under-agarose assay (bottom panel). In the micropipette assay, the micropipette (indicated by the arrow) was not moved for the duration of the 13 min video. Note the lack of translocation of the R42C-CRAC to the side of the cell facing the chemoattractant source. The under-agarose assay was carried out for extended times (2 hr) to verify the chemotaxis defective phenotype of R42C-CRAC. With this extended duration, wild-type cells form very long streams of cells migrating toward the chemoattractant source. For the biochemical translocation assay (middle panel), cells were stimulated with 10 μ M cAMP and membrane preparations analyzed as described in the legend of Figure 2. The results are representative of at least three independent experiments.

(C) Adenylyl cyclase activity in CRAC PH domain mutants. Differentiated cells were stimulated with cAMP and processed for adenylyl cyclase activity measurements as described in the legend of Figure 2. The graph is representative of at least six independent experiments, each performed in duplicate.

(D) Myr-CRAC is degraded upon the initiation of differentiation. Images show the distribution and presence of Myr-CRAC-GFP expressed in *crac*⁻ cells at different times after the initiation of differentiation. A larger field is presented for the 120 min time showing a larger number of cells, indicating a widespread loss of Myr-CRAC-GFP (phase image on right). The bottom panel shows Western blot detection of Myr-CRAC-GFP at various time points after the start of differentiation.

tions. First, they establish that PI3K activity is required for maximal chemoattractant-mediated stimulation of adenylyl cyclase, as observed for relaxin-mediated adenylyl cyclase activation in a monocyte cell line [17]. Second, our findings show that the CRAC determinants that independently regulate chemotaxis and adenylyl cyclase activation are downstream of PI3K.

In order to further assess the requirements for the membrane recruitment of CRAC for adenylyl cyclase activation, we fused CRAC to the *src* myristoylation (Myr) sequence, which has previously been used for targeting proteins to the plasma membrane [18]. This Myr-CRAC fusion protein showed constitutive membrane localization in undifferentiated cells. Remarkably, we found that the Myr-CRAC protein is rapidly internal-

ized and degraded upon the initiation of differentiation (Figure 3D), therefore precluding us from assessing the effects of constitutive membrane targeting of CRAC in chemotaxis-competent cells. Nevertheless, these results raise the interesting possibility that the recruitment of CRAC to the plasma membrane initiates a program of regulated CRAC turnover.

Overexpression of CRAC and CRAC Mutants Alters Chemoattractant Signaling

We reasoned that, because CRAC is important for multiple aspects of chemoattractant signaling, the overexpression of CRAC in wild-type Ax3 cells could also affect a variety of signaling pathways. We found that the overexpression of wild-type CRAC dramatically increases

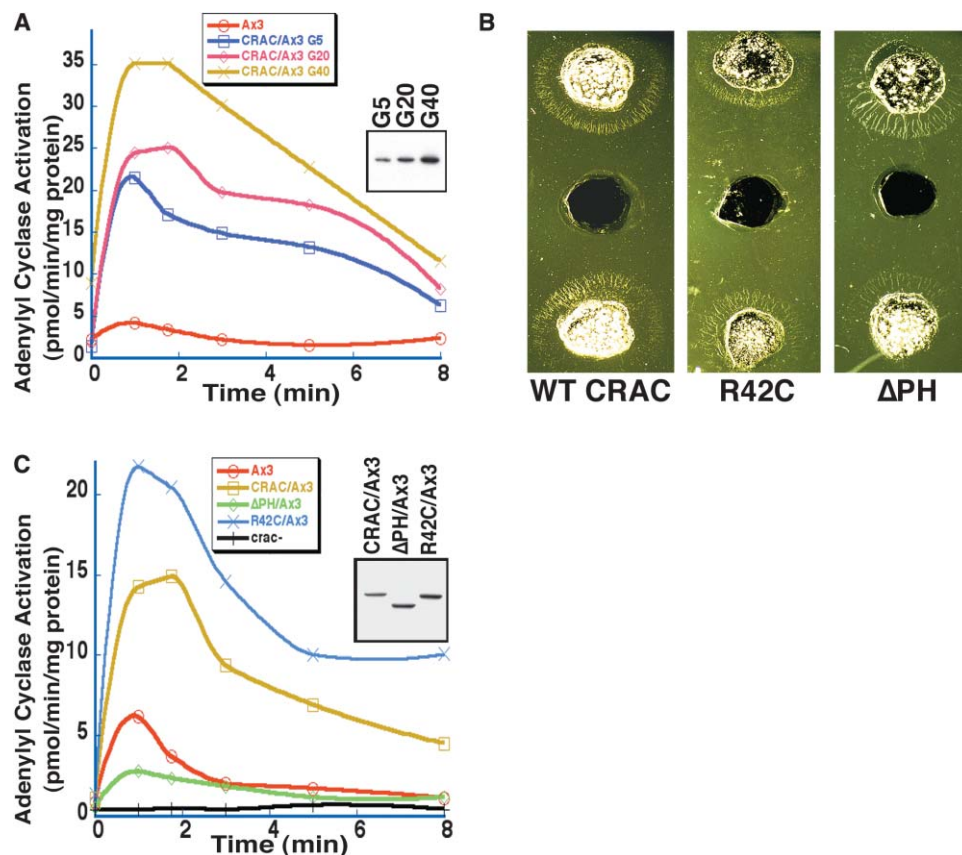


Figure 4. Overexpression of CRAC and CRAC Mutants Alters Chemoattractant Signaling to Adenylyl Cyclase

(A) Adenylyl cyclase activity in wild-type Ax3 cells expressing wild-type CRAC. To allow us to visually monitor the expression of our constructs, we used GFP fusions for these experiments. We overexpressed CRAC in wild-type cells, selecting for transformants at G418 concentrations of 5, 20, and 40 μ g/ml. Cells were assayed for chemoattractant-stimulated adenylyl cyclase activation as described in the legend of Figure 2. The inset shows an anti-GFP Western blot demonstrating the relative expression level of CRAC-GFP at each dose of G418.

(B) Under-agarose chemotaxis assays of wild-type cells expressing CRAC and CRAC PH mutant constructs. These transformants were generated using 40 μ g/ml G418. In each case, the untransfected wild-type cells are in the top well and the transfected cells are in the bottom well. Each experiment was performed at least three times in triplicate.

(C) Adenylyl cyclase activity in wild-type cells expressing the indicated CRAC constructs. Cells were assayed for chemoattractant-stimulated adenylyl cyclase activation as described in the legend of Figure 2. The inset shows an anti-GFP Western blot demonstrating the relative expression level of each construct.

the extent of adenylyl cyclase activation in response to chemoattractant (Figure 4A). Interestingly, we also find that CRAC overexpression does not have an apparent effect on chemotaxis (Figure 4B). We conclude that CRAC or a CRAC-interacting factor is limiting for adenylyl cyclase activation but not for chemotaxis. These results are consistent with our previously reported studies showing that ACA is not required for chemotaxis [14].

To further characterize the CRAC PH domain, we overexpressed various PH domain mutants in wild-type Ax3 cells and analyzed CRAC-dependent processes. Like wild-type CRAC, none of the overexpressed PH-domain mutants has a dramatic effect on chemotaxis, although cells expressing Δ PH-CRAC routinely display less robust movement and PH_{CRAC} expressed in wild-type cells have polarity defects (Figure 4B and data not shown). On the other hand, overexpression of some PH domain mutants leads to interesting effects on chemoattractant-mediated activation of adenylyl cyclase. While the overexpression of the PH domain alone has no major conse-

quence (data not shown), the overexpression of the R42C-CRAC has a dominant positive effect (Figure 4C) and the Δ PH-CRAC mutant produced the opposite effect, showing a dominant negative suppression of adenylyl cyclase activation (Figure 4C). From these results, we infer that adenylyl cyclase regulation requires the interaction of multiple factors with both the C-terminal region and the N-terminal PH domain, independent of the 3-phosphoinositide binding site. As shown for a variety of other PH domain-containing proteins, these observations also imply that, in addition to binding to lipid products of PI3K, the PH domain takes part in other interactions that are critical for CRAC function [19]. We propose that the dominant positive effect of overexpressed R42C-CRAC involves its binding to multiple factors that target and activate the adenylyl cyclase signaling complex. In contrast, the dominant negative effect of overexpressed Δ PH-CRAC involves its binding and sequestration or inactivation of essential factors that interact with the C-terminal region. We suggest that the

translocation of the CRAC complex initiates a series of events that ultimately lead to the full activation of ACA. In support of this hypothesis, we show here that R42C-CRAC, which fails to bind to 3-phosphoinositides and displays no measurable plasma membrane association, confers minimal ACA activation when expressed in *crac*[−] cells. In addition, *pten*[−] cells, which show an exaggerated and prolonged duration of CRAC translocation, display sharply increased chemoattractant-dependent activation of ACA [20]. Taken together, these results suggest that CRAC is a central regulator that mediates the action of multiple effectors of chemoattractant signaling.

Supplemental Data

Supplemental Data including Experimental Procedures, four additional figures, and four movies are available at <http://www.current-biology.com/cgi/content/full/15/2/134/DC1/>.

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